SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **R.DATI PRAVALLIKA/(192525473)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R.PRAVALLIKA

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

COT-2 ASS-1

① Suppose that the data for analysis includes the attributes age. The age Values for the data tuples 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

a) what is Mean and Median?

$$\boxed{\text{Mean}\ (\bar{x}) = \sum_{i=1}^{n} \frac{w_i x_i}{w_i}} \qquad \boxed{\frac{1}{N} \sum_{i=1}^{n} \frac{w_i x_i}{w_i}}$$

$$= \frac{809}{27} = 29.96 = 29.96 = 30.$$

$$\boxed{\text{Median}\ \left(N + \frac{1}{2}\right)} \quad 27 + \frac{1}{2} = 14$$

(b) Mode and Modality

25 and 35 are repeating for 4 times

it is biModel.

(c) Mid range

$$\begin{array}{r} 70 \\ 13 \\ \hline 83 \\ \hline \end{array}$$

$$\text{Mid rang:-}\ \frac{max + min}{2} = \frac{13 + 70}{2}$$

$$= 41.5$$

d, first Quantile and third Quantile.

Q_1 and Q_3

Q_1 = before middle values.

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

Midvalue = 20 = 7

$\boxed{Q_1 = 20}$

35
20
15

Q_3 = 30, 33, 33, 35, 35, 35, 36, 40,

45, 46, 52, 70.

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Midvalue = 35

$\boxed{Q_3 = 35}$

$\boxed{IQR = 15}$

$\boxed{Q_3 - Q_1}$

e, five-number summary

min, Q_1, median, Q_3, max

(13, 20, 25, 35, 70)

Min = 13

Q_1 = 20

median = 25

Q_3 = 35

max = 70

Boxplot of the data.

$Q_1 = 20, \ Q_3 = 35$

13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35,

36, 40, 45, 46, 52, 70.

13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35, 36, 40, 45, 46, 52, 70

② Given the following Measurements for the Variable age:

$\frac{1}{N} \sum_{i=1}^{n} \frac{w_i x_i}{w_i}$

18, 22, 25, 42, 28, 43, 33, 35, 56, 28,

Standarize the Variable by the following

Compute the Mean absoute deviation Of age.

$\text{Mean} = \frac{330}{10} = 33$

Age	$(x - \bar{x})$
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5

$MAD = \frac{88}{10} = 8.8$

③ Suppose that data for analysis includes the attribute age. The age values for the data tuples are (increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

depth of 3

Total value = 27.

no. bin = 3.

depth → 9

Total no. of bins = $\frac{27}{3}$ = 9 bins required.

Bins	Orginal values	Bin Mean	Smoothed value
B_1	13, 15, 16	14.67	14.67, 14.67, 14.67
B_2	16, 19, 20	18.33	18.33, 18.33, 18.33
B_3	20, 21, 22	21	21, 21, 21
B_4	22, 25, 25	24	24, 24, 24
B_5	25, 25, 30	26.67	26.67, 26.67, 26.67
B_6	33, 33, 35	33.67	33.67, 33.67, 33.67
B_7	35, 35, 35	35	35, 35, 35
B_8	36, 40, 45	40.33	40.33, 40.33, 40.33
B_9	46, 52, 70	56	56, 56, 56

(4) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalize by setting min=0 and max=1 (b) Z-score normalization

a) 200, 300, 400, 600, 1000

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Min = 200

Max = 1000

Value	Normalized
200	= 0.00
300	$\frac{300-200}{800} = 0.125$
400	$\frac{400-200}{800} = 0.25$
600	$\frac{600-200}{800} = 0.50$
1000	$\frac{1000-200}{800} = 1.00$

(b) $z = \frac{x-\mu}{\sigma}$

$$\mu = \frac{200+300+400+600+1000}{5} = 500$$

$\sigma =$

Value	$(x-\mu)$	$(x-\mu)^2$	z-s...
200	-300	90,000	
300	-200	40,000	
400	-100	10,000	
600	100	10,000	
1000	500	250,000	

$$\Sigma(x-\mu)^2 = 400000$$

$$\sigma^2 = \frac{400000}{5} = 80000$$

$$\sigma = \sqrt{80000} \simeq 282.84$$

$(200-500)/282.84 = -1.06$

$(300-500)/282.84 = -0.71$

$(400-500)/28.284 = -0.35$

$(600-500)/282.84 = 0.35$

$(1000-500)/282.84 = 1.77$

$-1.06, -0.71, -0.35, 0.35, 1.77$